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Virtual Mass Multiphase Flow Meter (vMPFM) – A Digital Future Enabled by AI/ML & PINN

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Abstract

As the energy industry transitions toward digitalization and environmental accountability, real-time and accurate flow measurement in multiphase environments has become essential for production optimization, emissions management and leak detection. This paper introduces the Virtual Multiphase Mass Flow Meter (vMPFM), an AI/ML and PINN-enabled multiphase flow metering system that eliminates the need for complex hardware installations by utilizing existing field sensors, and demonstrates how the integration of PINNs, trained and validated using CFD-simulated environments, can enable accurate, scalable, and real-time mass flow measurement together with leak detection and emissions measurement across field-wide deployments, including remote wellheads and processing facilities.

The proposed vMPFM architecture leverages existing field sensor data available at the wellhead or processing facilities to infer real-time multiphase mass flowrates. Using a hybrid modelling approach, CFD is conducted to model diverse flow regimes and sensor responses, forming the primary training and validation dataset for a PINN. Physical constraints - such as the Navier-Stokes equations, interfacial closure laws and Euler-Lagrange Model are embedded directly into the training process, enabling the model to learn flow behavior consistent with governing fluid dynamics. Calibration incorporates historical flow test data, flow loop benchmarks, and where feasible, field-deployed measurements. This approach ensures robustness and adaptability to a wide range of operating conditions. The trained model is deployed as a vMPFM for field wide monitoring, either as a standalone digital solution or embedded in minimal edge computing devices (e.g. compact hybrid MPFM units) for remote operations. Furthermore, the applicability to use vMPFM as a leak detection tool was evaluated against traditional numerical methods.

Validation of the vMPFM across diversely simulated operational regimes demonstrates its robustness and high accuracy. Benchmarking against traditional MPFMs and flow loop shows that the vMPFM system consistently achieves mass flow measurement accuracy within 5%, even under varying flow conditions from low to high water cut, low to high Gas Volume Fraction (GVF), and even fugitive emissions events. The vMPFM adapts dynamically to changing flow profiles without the need for frequent recalibration, significantly lowering maintenance overhead and measurement uncertainty. By leveraging only conventional sensor inputs, it delivers continuous real-time mass flow data with minimal hardware intervention. These capabilities enable proactive production surveillance, early leak detection, and precise

emissions mass quantification. Furthermore, its scalable digital architecture extends measurement capacities to locations previously deemed impractical or cost-prohibitive to monitor, reducing both capital and operating expenditures while supporting broader field-wide digitalization, decarbonization and sustainable development goals.

This paper presents a novel framework for digitized multiphase mass flow measurement that unifies physical modeling and data-driven learning. By removing the dependency on costly hardware and enabling scalable deployment through existing field sensor, the vMPFM redefines how multiphase flows are measured, monitored and managed including potential applications of emissions monitoring and leak detection—bridging the gap between conventional metering and intelligent field development.

Introduction

The energy industry is undergoing a profound transformation as digitalization, sustainability, and operational efficiency become core drivers of upstream and midstream developments. This shift is particularly pronounced in the context of multiphase mass flow measurement, where accurate and real-time quantification of oil, gas, and water phases remains a cornerstone of production optimization, environmental accountability, and asset integrity management. In conventional production environments, multiphase flow measurement relies heavily on complex hardware setup—ranging from radioactive densitometers, electromagnetic/capacitance/conductivity sensors and differential pressure devices to expensive test separators—that often suffer from high capital expenditures (CAPEX), operational complexities, and limited scalability across large field deployments. These limitations become especially critical in remote, offshore, and unconventional reservoirs, where continuous monitoring become challenging with traditional instrumentation (Falcone et al., 2009; Pedersen et al., 2015).

To address these challenges, this paper introduces a Virtual Mass Multiphase Flow Meter (vMPFM), a data-driven yet physics-informed system architecture that utilizes existing field instrumentation to infer real-time multiphase flowrates without the need for hardware-intensive metering setups. Assuming adequate sensor coverage, vMPFM leverages existing SCADA data for scalability. The core innovation lies in the integration of Physics-Informed Neural Networks (PINNs) with computational fluid dynamics (CFD)-generated synthetic datasets, forming a hybrid modeling framework that unifies domain knowledge with machine learning for robust and generalizable flow estimation. The vMPFM offers a practical pathway toward scalable, intelligent flow monitoring that aligns with industry efforts to de-carbonize operations, reduce methane emissions, and enhance surveillance capabilities in real-time (International Energy Agency, 2021; Brown and Durdevic, 2020).

The Importance of Multiphase Mass Flow Measurement in Digital Oilfields

Multiphase mass flow occurs ubiquitously within oil and gas production systems, particularly throughout the journey from the reservoir to surface separation facilities. Accurate determination of mass flow rates for individual phases—oil, water, and gas—is critical for effective reservoir management, well testing, accurate production allocation, and compliance with stringent regulatory requirements on emissions reporting.

Traditional test separators, while historically reliable, typically rely on volumetric measurements and are inadequate for the continuous mass monitoring demands of modern digital oilfields. Their high operational costs, susceptibility to pressure and temperature variations, and inherently intermittent batch measurement approach further limit their efficacy for precise mass quantification (Jamaluddin et al., 2001). Conventional inline multiphase flow meters (MPFMs), which also commonly depend on volumetric assessments, offer real-time data but frequently face challenges related to sensitivity to varying flow regimes, calibration drift, and substantial maintenance demands (Alimonti and Savioli, 2016).

Mass flow measurements provide significant advantages over conventional volumetric measurements by accounting directly for variations in density and composition, eliminating the need for separate density

determination and reducing uncertainties associated with fluid property changes. This enhanced accuracy is essential for precise fiscal allocation, reservoir characterization, and production optimization. Moreover, in digital oilfields, where data-driven decisions rely heavily on accuracy and real-time responsiveness, mass-based measurements are inherently more reliable and suitable.

Furthermore, with increasingly stringent global environmental regulations—including the U.S. EPA's Subpart W methane reporting guidelines and Canada's Methane Regulation under the Environmental Protection Act—the precise measurement of multiphase mass flow rates has become a critical operational necessity. Accurate mass flow measurement is especially important during venting, flaring, and transient operational events, ensuring reliable monitoring and potential quantification for regulatory compliance, environmental stewardship, and proactive emissions management ([U.S. Environmental Protection Agency, 2023](#)).

Limitations of Conventional MPFM Technology

Despite advances in sensor technology and embedded computing, conventional MPFMs remain constrained by their dependence on physical installations and predefined empirical or semi-empirical correlations, often tailored to specific flow regimes. These devices typically measure phase volumes directly, subsequently calculating mass flow rates by applying assumed or periodically measured density values. This approach introduces additional uncertainties, particularly when fluid composition and densities vary due to changes in pressure, temperature, or phase interactions.

For example, the drift-flux model, employed in many commercial MPFMs, assumes steady-state, axisymmetric flow conditions and frequently requires calibration against external test separator data. Such models exhibit significant performance degradation under conditions of slugging or annular flow, and their accuracy diminishes notably in high water-cut or gas-dominated environments ([Franca and Lahey Jr, 1992](#); [Frøysa and Skjæveland, 2001](#)). Furthermore, the reliance on radioactive densitometers, commonly used for density determination in conventional MPFMs, presents additional safety and logistical challenges, demanding specialized handling, regulatory compliance, and restricting their practical scalability in field operations.

The Digital Shift: AI and Hybrid Modeling for Flow Inference

In recent years, the integration of artificial intelligence and machine learning (AI/ML) with physics-based modeling has enabled significant advancements in soft sensing for multiphase flow systems. Hybrid modeling strategies that blend first-principles physics with data-driven learning have demonstrated strong potential to reduce reliance on extensive empirical datasets while preserving physical interpretability and adherence to conservation laws ([Karniadakis et al., 2021](#); [Raissi et al., 2019](#)). Among these, Physics-Informed Neural Networks (PINNs) have gained attention as a particularly robust framework, as they embed governing equations—such as the Navier-Stokes equations, mass and momentum conservation laws, and interfacial interaction models—directly into the training process. Notably, PINNs can infer key fluid properties such as density dynamically, without requiring direct sensor measurements. This inherent capability supports the digital scalability of PINN-based systems like the vMPFM, enabling accurate multiphase flow estimation even in sparsely instrumented environments.

Unlike traditional neural networks that rely solely on input-output pairs, PINNs minimize a composite loss function consisting of data-driven and physics-based residuals, allowing them to generalize better in sparse-data regimes and extrapolate across varying operating conditions ([Mao et al., 2020](#)).

The Role of CFD in Training and Validation

The use of CFD as a digital twin platform to generate high-fidelity training and validation datasets is a critical enabler for the proposed vMPFM system. CFD simulations capture complex flow phenomena such as phase slip, interfacial drag, turbulence, and geometric constraints under controlled boundary conditions.

This allows for the generation of synthetic datasets that span a wide range of flow rates, compositions, and flow regimes—conditions that would be difficult or expensive to reproduce in physical flow loops or field tests (ANSYS Inc., 2024).

The CFD models are constructed using multiphase Navier-Stokes solvers under the Eulerian-Eulerian and Euler-Lagrangian formulations, with interfacial closure models such as the Schiller-Naumann drag law and population balance models where applicable (Versteeg and Malalasekera, 2007).

vMPFM Architecture and Deployment Strategy

The vMPFM system architecture consists of a modular pipe that includes: (1) sensor signal preprocessing, (2) real-time inference using a trained PINN model, and (3) optional calibration and feedback integration using historical test separator or flow loop data. Input features may include standard wellhead measurements such as tubing pressure, casing pressure, temperature, differential pressure (across choke or venturi), and acoustic signatures—all of which are typically available from existing SCADA systems.

Once trained, the model can be deployed as a software layer integrated into existing control systems or embedded in additional physical edge computing devices for remote operations. This enables scalable deployment across a wide array of field assets, including marginal wells, unmanned platforms, and gathering systems (Gao et al., 2022). Importantly, the vMPFM requires no radioactive sources or complex installations, drastically reducing the barriers to deployment.

Leak Detection and Emissions Monitoring Applications

Beyond production monitoring, the vMPFM enables new capabilities in leak detection and emissions quantification. By continuously comparing expected flow behavior with real-time sensor inputs, the system can detect deviations indicative of leaks, restrictions, or equipment malfunctions. This can be further enhanced through integration with wind data, GPS-tracked gas dispersion models, or secondary field sensors such as infrared cameras, which will be the subject of future studies.

In emissions measurement applications, the vMPFM provides real-time mass flow estimates for gas discharge events, facilitating accurate greenhouse gas reporting and operational interventions during blowdowns, flaring, or venting (Zhang et al., 2020; CSA Group, 2021).

Performance Benchmarking and Validation

To validate its performance, the vMPFM was benchmarked against traditional MPFMs under controlled flow loop conditions and simulated transient scenarios. The results consistently showed that vMPFM maintains measurement accuracy within $\pm 5\%$ across a broad range of gas volume fractions (GVF), water cuts, and total flow rates. Notably, the system remained stable during flow regime transitions and under unsteady conditions such as slugging and fugitive emissions events—scenarios where traditional MPFMs often exhibit degraded accuracy or require recalibration.

Implications for Intelligent Field Development

The implications of vMPFM extend beyond metering into the broader domain of intelligent field development. The system supports continuous, autonomous monitoring at a fraction of the cost of traditional infrastructure, enabling field-wide data integration, early anomaly detection, and predictive maintenance strategies. This supports the development of "smart wells" and "digital twins," fostering a transition toward fully autonomous, low-carbon oil and gas operations.

By lowering the entry barrier for real-time measurement, the vMPFM framework also democratizes access to advanced surveillance tools in low-margin or developing assets, unlocking new economic and operational value. Its architecture aligns with the broader goals of energy transition, including methane reduction, electrification, and digital integration across upstream infrastructure (BP, 2021; Schlumberger, 2023).

This paper specifically addresses these industry challenges by presenting the Virtual Mass Multiphase Flow Meter (vMPFM), designed with the explicit objectives of achieving accurate real-time multiphase mass flow measurement, enhancing emissions quantification capabilities, and enabling early and reliable leak detection across diverse and challenging operational scenarios.

Theory

Multiphase flow systems, especially those prevalent in oil and gas production environments, involve simultaneous flow of immiscible phases—typically oil, water, and gas—through complex geometries such as pipes and wellbores. Modeling these flows accurately is essential for flow metering, production optimization, and reservoir management. However, traditional mathematical models often rely on simplifying assumptions and empirical correlations, limiting their adaptability and generalization. In this section, we present the governing physics of multiphase mass flow, the closure relations necessary for solving these equations, and the formulation of Physics-Informed Neural Networks (PINNs) that integrate this physics into a machine learning framework.

Fundamentals of Multiphase Flow Physics

Multiphase flow is governed by a set of conservation equations derived from the fundamental principles of mass, momentum, and energy conservation. These equations are typically expressed in an Eulerian framework, where each phase is treated as a continuous medium with its own velocity and pressure fields (M. Ishii, 1975). The general form includes phase continuity and momentum equations.

Each fluid phase $k \in \{\text{oil, water, gas}\}$ is described by its own continuity equation:

$$\frac{\partial}{\partial t}(\alpha_k \rho_k) + \nabla \cdot (\alpha_k \rho_k \mathbf{u}_k) = \Gamma_k \quad (1)$$

where α_k is the volume fraction, ρ_k the density, $(\mathbf{u}_k)$ the velocity, and Γ_k the mass transfer rate of phase k .

The volume fractions must satisfy:

$$\sum_k \alpha_k = 1 \quad (2)$$

Phase Momentum Equation. The momentum conservation equation for phase k accounts for pressure gradients ∇p , viscous forces, gravity $\mathbf{g}$, and interfacial forces:

$$\frac{\partial}{\partial t}(\alpha_k \rho_k \mathbf{u}_k) + \nabla \cdot (\alpha_k \rho_k \mathbf{u}_k \otimes \mathbf{u}_k) = -\alpha_k \nabla p + \nabla \cdot (\alpha_k \boldsymbol{\tau}_k) + \alpha_k \rho_k \mathbf{g} + \mathbf{M}_k \quad (3)$$

where $\boldsymbol{\tau}_k = \mu_k (\nabla \mathbf{u}_k + \nabla \mathbf{u}_k^\top)$ is the viscous stress tensor, and μ_k represents dynamic viscosity. $\mathbf{M}_k$ represents the interfacial momentum exchange term (M. Ishii, 1975).

Mixture Model Approximation. To reduce model complexity, the mixture model assumes a shared velocity and pressure field for all phases:

$$\partial \rho_m \partial t + \nabla \cdot (\rho_m \mathbf{u}_m) = 0 \quad (4)$$

$$\partial (\rho_m \mathbf{u}_m) \partial t + \nabla \cdot (\rho_m \mathbf{u}_m \otimes \mathbf{u}_m) = -\nabla p + \nabla \cdot (\mu_m \nabla \mathbf{u}_m) + \rho_m \mathbf{g} \quad (5)$$

with:

$$\rho_m = \sum_k \alpha_k \rho_k, \quad \mu_m = \sum_k \alpha_k \mu_k \quad (6)$$

This model works well for homogeneous or dispersed flows and is widely used in industrial simulation tools such as OLGA and ANSYS Fluent (M. Ishii, 2006).

However, under more complex flow conditions involving phase slip, the Mixture Model may fail to accurately represent the flow behavior, necessitating the use of alternative models such as the Drift-Flux Model.

Drift-Flux Model. When significant phase slip exists (e.g., gas bubbles in liquid), the drift-flux model captures velocity differences:

$$\mathbf{u}_g = \mathbf{u}_m + \mathbf{u}_d \quad (7)$$

$$j_g = \alpha_g \mathbf{u}_g = C_0 \alpha_g \mathbf{u}_m + \alpha_g V_{gj} \quad (8)$$

where C_0 is the distribution parameter and V_{gj} is the drift velocity (N. Zuber and J.A. Findlay, 1965).

Closure Relations

Interfacial Momentum Transfer. Closure for interfacial momentum includes drag, lift, and virtual mass forces:

$$\mathbf{F}_{da,k} = \sum_{l \neq k} K_{kl} (\mathbf{u}_l - \mathbf{u}_k) \quad (9)$$

with:

$$K_{kl} = \frac{3C_D \alpha_k \alpha_l \rho_m}{4d_b} |\mathbf{u}_l - \mathbf{u}_k| \quad (10)$$

where C_D is the drag coefficient and d_b is the droplet or bubble diameter, α_l is the liquid fraction (C.T. et al., 1998).

Volume Fraction Transport. The transport equation for α_k is:

$$\frac{\partial \alpha_k}{\partial t} + \nabla \cdot (\alpha_k \mathbf{u}_k) = S_k \quad (11)$$

S_k includes source terms due to coalescence, breakup, or interphase mass transfer (C.E. Brennen, 2005).

Mixture Properties. Effective properties are modeled via weighted averages:

$$\rho_{mixture} = \sum_k \alpha_k \rho_k, \quad \mu_{mixture} = \sum_k \alpha_k \mu_k \quad (12)$$

In emulsions, nonlinear models may be used:

$$\mu_{emulsion} = \mu_{oil} (1 + 2.5\alpha_w + 10.05\alpha_w^2 + \dots) \quad (13)$$

Turbulence and Boundary Conditions. Turbulence is often treated using mixture-based $k - \epsilon$ or LES models. Boundary conditions include:

Inlet: Prescribed velocity or flow rates.

Outlet: Fixed pressure or zero-gradient conditions.

Wall: No-slip for liquids; possibly slip for gas.

Physics-Informed Neural Networks (PINNs) for Multiphase Navier-Stokes Equations

PINNs combine deep neural networks with partial differential equation (PDE) constraints to learn physical solutions from sparse or noisy data [9]. In multiphase applications, PINNs derive mass flow rates by solving continuity and momentum equations while dynamically inferring fluid densities in real time. The neural network, denoted as $f_\theta(\mathbf{x}, t)$, is trained to approximate the underlying physical fields across space and time:

$$f_\theta(\mathbf{x}, t) \rightarrow [\mathbf{u}_m, p, \alpha_k] \quad (14)$$

p is the pressure distribution. PINNs use computational techniques - automatic differentiation to evaluate derivatives for PDE residuals, enabling efficient computation of PDE constraints.

Loss Function. The total loss includes sensor measurement data, physics, and boundary/initial condition components:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \mathcal{L}_{\text{physics}} + \mathcal{L}_{\text{BC/IC}} \quad (15)$$

$\mathcal{L}_{\text{data}}$: Fit to measured sensor data (e.g., pressure, differential pressure, temperature, etc.)

$\mathcal{L}_{\text{physics}}$: Residuals of N-S and continuity equations or other governing equations.

$\mathcal{L}_{\text{BC/IC}}$: Residuals for boundary and initial condition.

SR-RK-PINN. The Self-Regressive Runge-Kutta Physics-Informed Neural Network (SR-RK-PINN) was chosen for its efficiency in modeling transient multiphase flow, as demonstrated by Wei et al. (2024). This method excels in dynamic environments by integrating Runge-Kutta time-stepping with a self-regressive training mechanism, enabling accurate simulation of complex multiphase interactions.

The core architecture of the SR-RK-PINN combines the strengths of numerical time integration and self-supervised learning. The model begins by training on initial conditions and boundary data, then predicts the solution for the next time step. These predictions are iteratively incorporated as new training labels, enabling the model to dynamically evolve over time. This recursive, multi-step prediction loop significantly improves temporal generalization and accuracy when solving partial differential equations (PDEs).

To further enhance training efficiency and accuracy, the optimization process employs a hybrid strategy. It uses the Adaptive Moment Estimation (Adam) algorithm for global parameter exploration—thanks to its adaptive learning rate and gradient moment tracking—followed by the Limited-memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS) algorithm, a quasi-Newton method effective in local convergence and fine-tuning. This two-stage approach reduces computational cost and ensures stable, high-precision solutions.

This methodology has been successfully applied to solve nonlinear PDEs such as the Allen-Cahn and Burgers' Equations, as reported by Wei et al. (2024), validating its robustness and adaptability. The computational workflow is illustrated in Figure 1, and the training process is outlined in Figure 2.

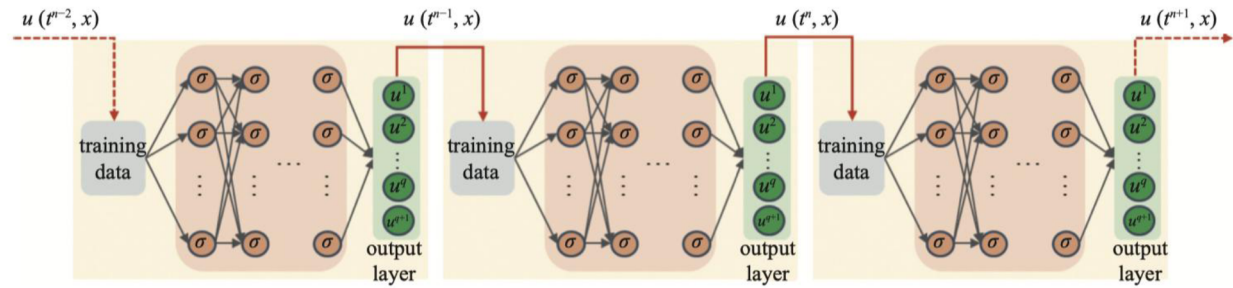


Figure 1—Schematic Diagram of the Structure of the SR-PINN-RK model

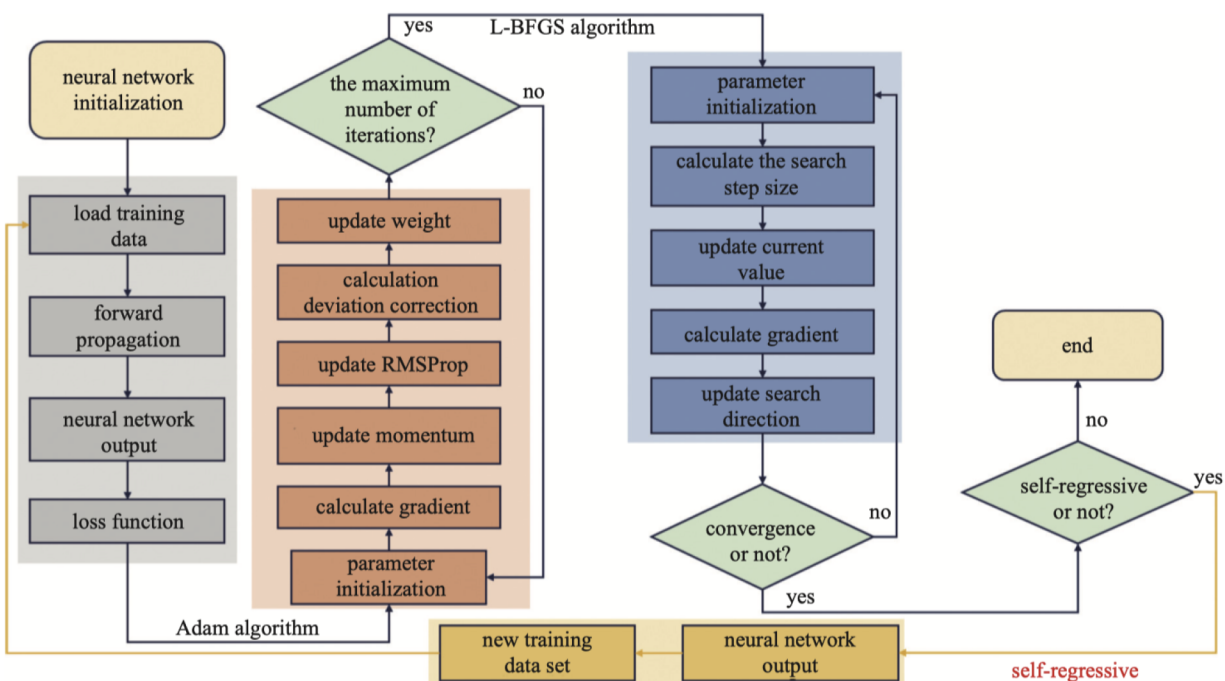


Figure 2—The Training Process Flowchart of the SR-PINN-RK model

The integration of these theoretical principles within the PINN framework directly enables the vMPFM to achieve high measurement accuracy, adaptability, and robustness across various flow regimes, as demonstrated in subsequent validation and results sections.

Methodology

To comprehensively validate the vMPFM's capabilities outlined in the Abstract—accurate multiphase mass flow measurement, emissions monitoring, and leak detection—three distinct experimental tracks were established, each specifically targeting one of these critical operational objectives. To evaluate the performance and applicability of the proposed virtual multiphase mass flow meter (vMPFM), a structured testing campaign was developed comprising three experimental tracks:

1. vMPFM vs. Flow Loop Benchmarking
2. vMPFM Application in Emissions Monitoring
3. vMPFM Application in Leak Detection

Each track targets a specific operational requirement of the vMPFM and is designed to assess its accuracy, robustness, and real-time responsiveness under varying process conditions. The methodology for each test track is detailed below.

vMPFM vs. Flow Loop Benchmarking

The experimental validation of the vMPFM was conducted at a full-scale, closed-loop multiphase flow test facility (flow loop) specifically designed to replicate the complex flow regimes encountered in upstream oil and gas production systems. The flow loop allows for controlled testing of three-phase mixtures—oil, water, and gas—across a wide range of flow conditions, enabling detailed benchmarking of metering technologies under near-field operational environments. The system operates at pressures up to 800 psi and temperatures up to 180°C, accommodating flow rates from 100 to 3,000 barrels per day (BPD) for liquids and up to 3 million standard cubic feet per day (MMSCFD) for gas. The test loop includes configurable piping

geometries (horizontal, vertical, and inclined) and is instrumented with sensors for pressure, temperature, phase fraction, and reference flow rates to ensure precise test conditions and repeatability.

The evaluation focused on three critical operational variables: water cut, gas volume fraction (GVF), and total mass flow rate. Water cut was varied from 0% to 100% in 20% increments to simulate a full range of liquid phase compositions, from pure oil to pure water. GVF was adjusted from 10% to 90%, capturing diverse gas-liquid flow regimes such as bubbly, slug, and annular flow. Total mass flow rate was tested from 1,000 kg/h to 12,000 kg/h to reflect both low- and high-throughput field scenarios. At each test condition, the vMPFM predictions were compared to reference values obtained from a test separator and benchmarked against a commercial mathematical model-based MPFM. This approach ensured a rigorous, multi-regime validation of vMPFM under realistic multiphase conditions.

vMPFM Application in Emissions Monitoring

To validate the ability of the virtual multiphase mass flow meter (vMPFM) to quantify potential emissions in field conditions, a test configuration was established involving both distributed vMPFM and centralized physical MPFM installations. Specifically, multiple vMPFMs were deployed at the wellhead level on a multi-well pad (WH-vMPFM), each responsible for measuring the individual well's multiphase production. In parallel, a single MPFM (GS-MPFM) was installed at the gathering station to measure the commingled flow after all individual streams had been combined. Emissions were simulated via deliberate gas releases between wellheads and the centralized gathering station.

The primary objective of this configuration was to assess whether the difference between the totalized gas production measured at the wellheads by vMPFMs and the commingled gas flow measured downstream could be used to detect unaccounted non-routine emissions, such as venting, fugitive leaks, or flare releases. By comparing the aggregated wellhead outputs to the centralized measurement at the gathering point, this approach aimed to assess the effectiveness of the vMPFM in enabling early detection and approximate ("ballpark") quantification of emissions in real time. Future studies will incorporate analyses of additional sources of measurement discrepancies or pipeline losses, such as pressure drops or phase separation effects.

vMPFM Application in Leak Detection

A controlled experiment was conducted to evaluate the capability of a virtual multiphase mass flow meter (vMPFM) system to detect and estimate small leaks under well-defined conditions. The test setup included two vMPFM units—one positioned upstream and the other downstream—across a 20-meter section of 3-inch diameter horizontal pipe. A ½-inch port with a ball valve was installed at the midpoint of the pipe to simulate a leak.

This port was connected to a calibrated turbine flow meter, which served as a reference for the actual leak rate. By analyzing the difference in flow measurements between the upstream and downstream vMPFMs, the test aimed to assess the system's ability to:

- Detect small-scale leaks,
- Quantify leakage rates in real time, and
- Evaluate the potential of vMPFM-based leak detection monitoring against a physical benchmark.

Results

The following results explicitly validate the performance claims presented in the Abstract, demonstrating the VMPFM's capability to consistently achieve $\pm 5\%$ measurement accuracy across diverse flow regimes, robustly monitor and estimate emissions rates during controlled release events, and reliably detect and measure leak events using distributed mass-balance methodologies.

vMPFM vs. Flow Loop Benchmarking

To evaluate the performance of the proposed Physics-Informed Neural Networks (PINNs) within the virtual multiphase mass flow meter (vMPFM) framework, its predictions were compared against those from a traditional mathematical model-based multiphase flow meter (MPFM) and benchmark data obtained from flow loop tests. The evaluation covered three distinct scenarios: (1) varying water cut, (2) varying gas volume fraction (GVF), and (3) varying total mass flow rate. All reported errors represent comparisons between either the MPFM or vMPFM (PINN) predictions and the reference flow loop benchmark measurements.

Variation in Water Cut. Tests were conducted by varying water cut from 0% (pure oil) to 100% (pure water) in 20% increments, under three GVF conditions: 10%, 50%, and 90%. At low GVF (10%), the vMPFM (PINN) outperformed MPFM across all flow parameters. MPFM errors against the flow loop ranged from 3–6%, especially in oil and water rates due to assumptions like homogeneous mixing. In contrast, the vMPFM (PINN) maintained sub-2% errors against the same flow loop consistently by leveraging physical constraints. At medium GVF (50%), both models showed strong performance, benefiting from well-mixed conditions. MPFM errors were typically <3%, while the vMPFM (PINN) refined predictions to <1–2%. Under high GVF (90%), the MPFM showed notable errors (up to 12%) due to high gas slip and phase separation. The vMPFM (PINN) maintained 2–3% accuracy across all rates, illustrating its robustness in challenging annular/mist flows.

Low GVF (10%)

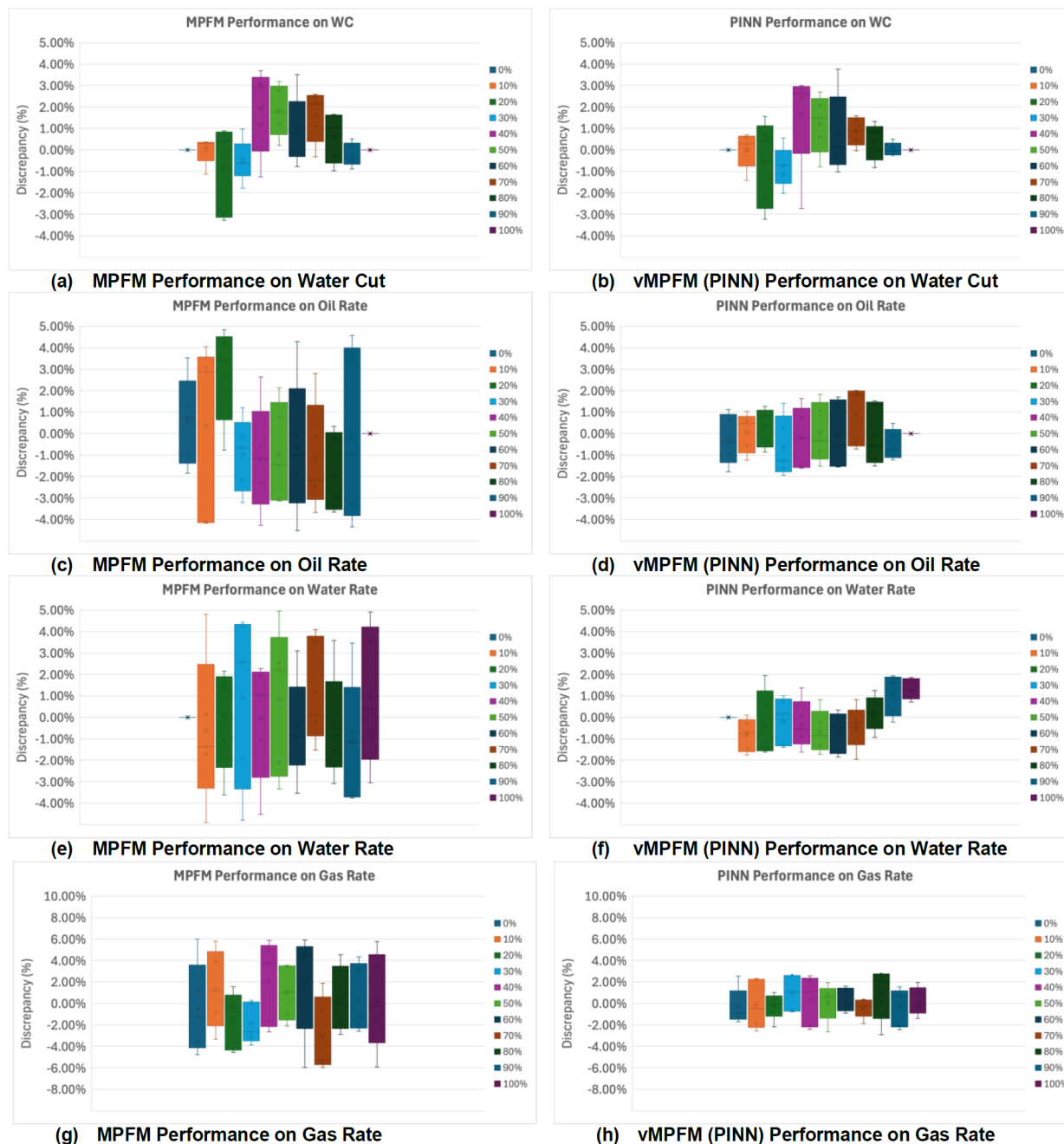


Figure 3—MPFM vs vMPFM (PINN) Performance on different WC on Low GVF (10%)

Medium GVF (50%)

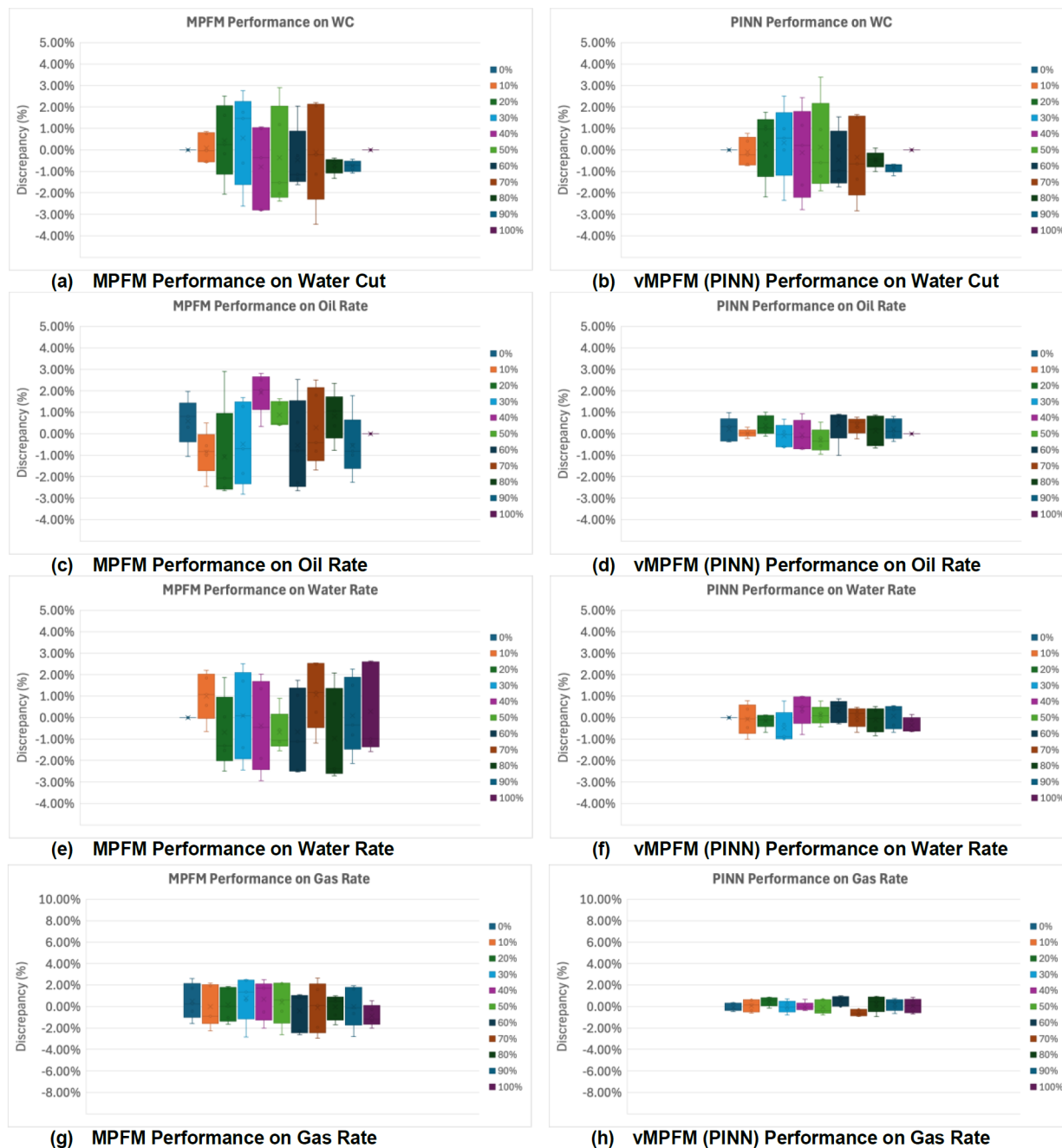


Figure 4—MPFM vs vMPFM (PINN) Performance on different WC on Medium GVF (50%)

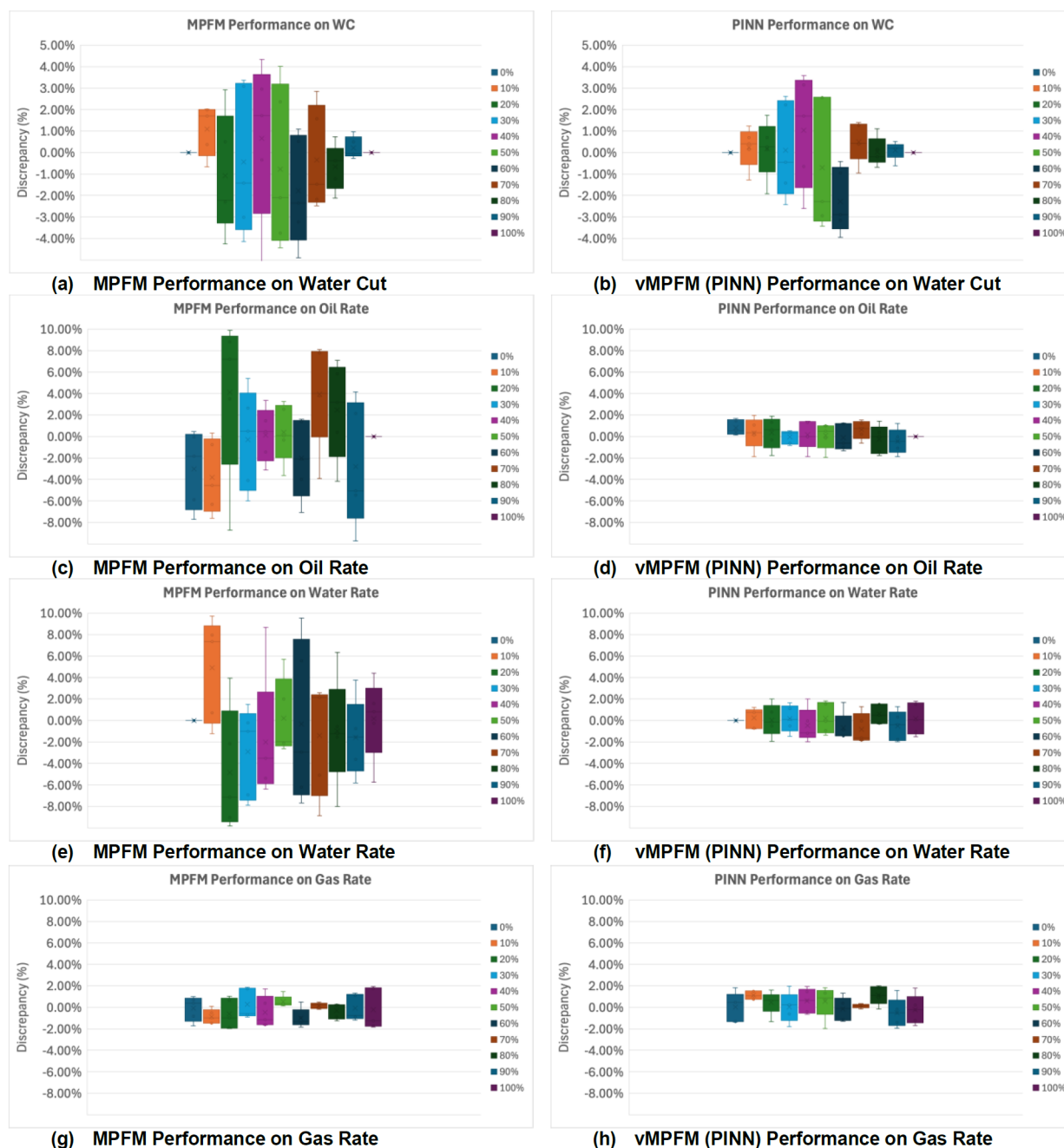
High GVF (90%)

Figure 5—MPFM vs vMPFM (PINN) Performance on different WC on High GVF (90%)

Variation in Gas Volume Fraction (GVF). GVF varied from 10% to 90% in 20% increment under three liquid-phase scenarios: pure oil, emulsion (50% water cut), and pure water. In the pure oil case, MPFM showed increasing error at higher GVF due to limitations in drift-flux assumptions, particularly with dispersed gas behavior. vMPFM (PINN) maintained stable oil and gas rate predictions with <2% error across the GVF range against the flow loop. In the emulsified (50% WC) scenario, three-phase flow complexity challenged the MPFM, which showed 3–6% errors in oil, water, and gas rates. The vMPFM (PINN) again performed better, maintaining 1–2% error, due to its data-driven learning ability to model nonlinear phase interactions without the traditional reliance on empirical regime maps by most conventional MPFMs. In the pure water scenario, the MPFM suffered from high errors (up to 6%) at high GVF due to gas-liquid slip. The vMPFM (PINN) remained accurate (<2.5%) in both gas and water rates.

Pure Oil (0% WC)

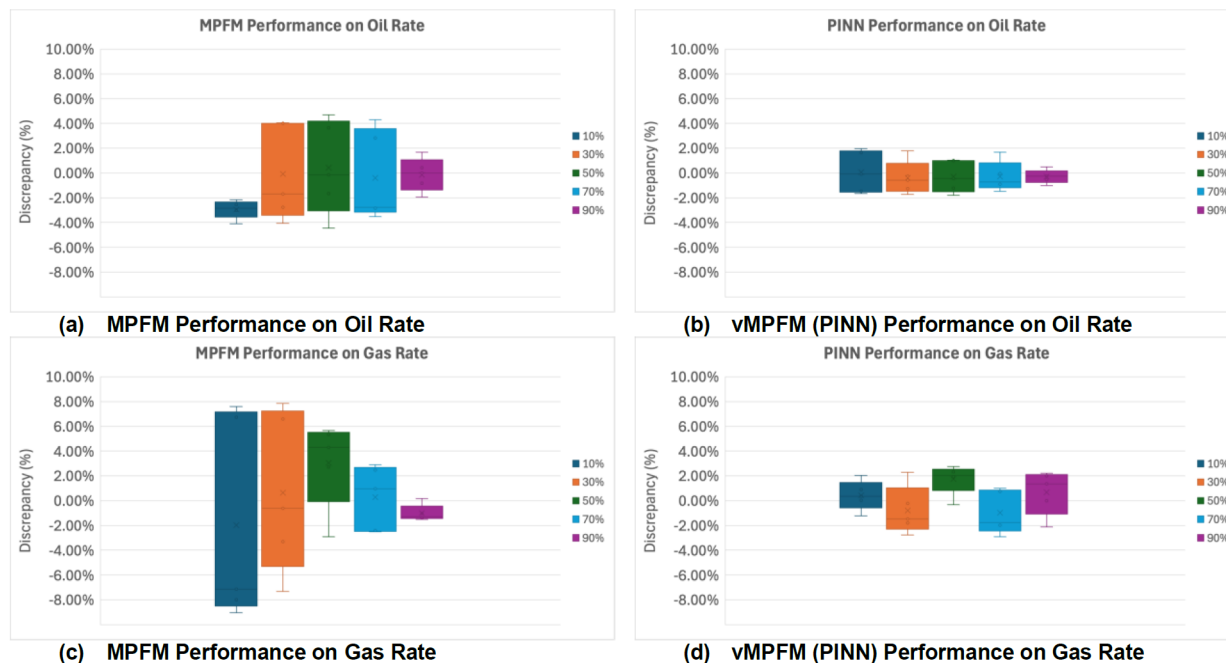


Figure 6—MPFM vs vMPFM (PINN) Performance on different GVF on Pure Oil (0% WC)

Emulsified flow (50% WC)

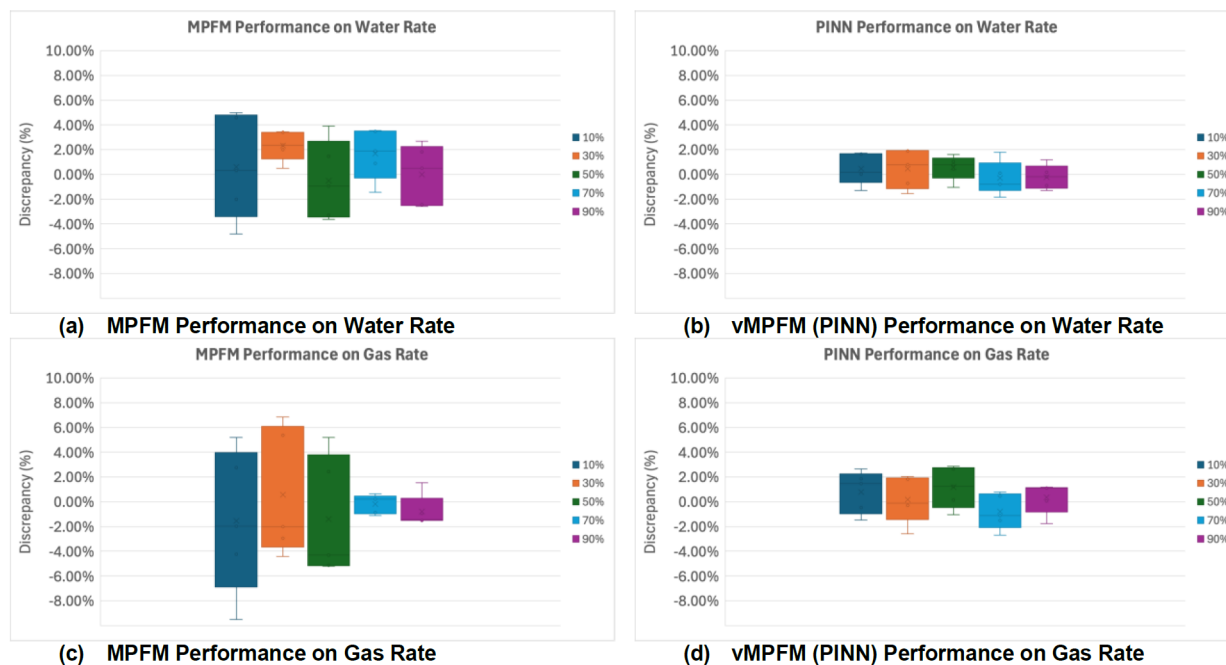


Figure 7—MPFM vs vMPFM (PINN) Performance on different GVF on Emulsified Flow (50% WC)

Pure Water (100% WC)

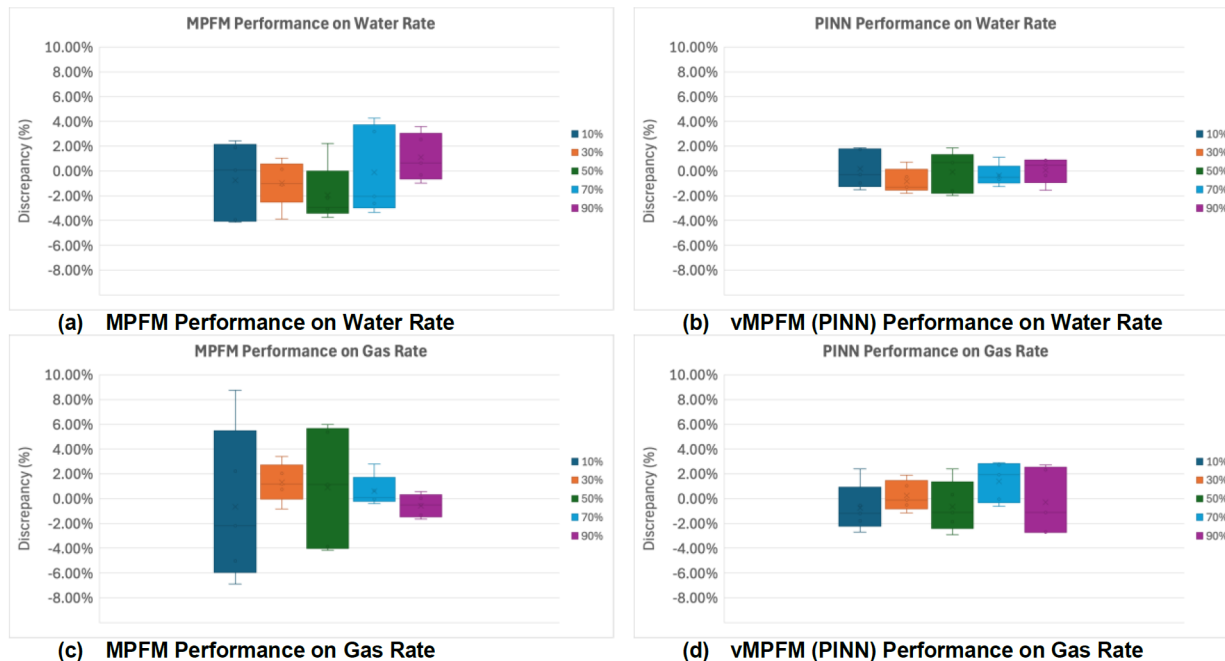


Figure 8—MPFM vs vMPFM (PINN) Performance on different GVF on Pure Water (100% WC)

Variation in Total Mass Flow Rate. Tests were conducted at flow rates ranging from 1,000 to 12,000 kg/h, with fixed conditions of 50% gas volume fraction (GVF) and 50% water cut (WC). These represent favorable conditions for MPFM performance due to stable mixing. Even under these optimal circumstances, traditional MPFM measurements exhibited errors of approximately 2–4% for oil, water, and gas rates compared to the flow loop benchmark. In contrast, the vMPFM employing Physics-Informed Neural Networks (PINNs) consistently achieved errors of less than 1–2%. Thus, under relatively stable flow regimes, the vMPFM (PINN) clearly outperformed the traditional MPFM, providing results in closer agreement with benchmark data.

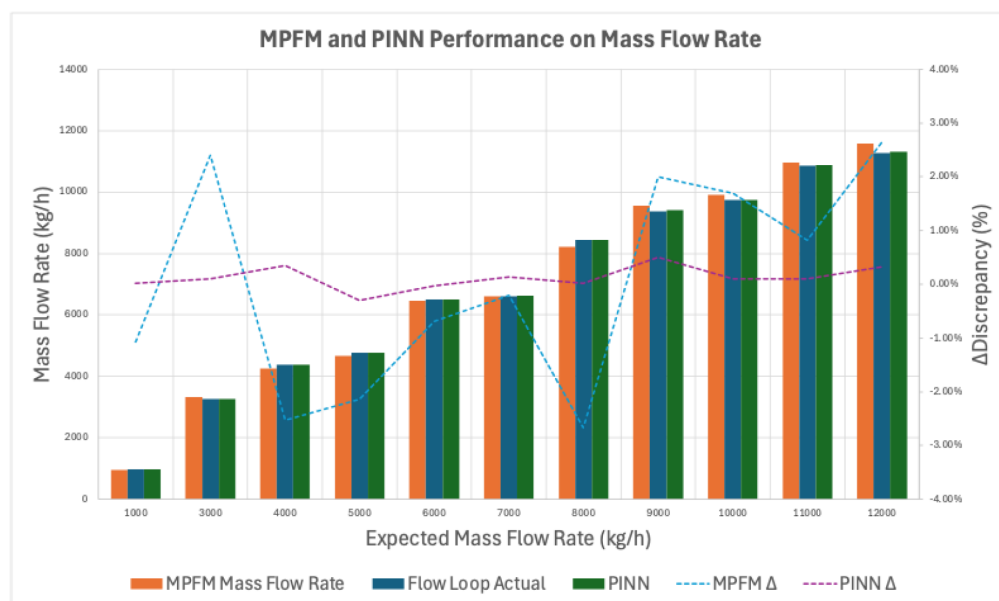


Figure 9—MPFM vs vMPFM (PINN) Performance on Different Mass Flow Rate @ 50% WC and 50% GVF

vMPFM Application in Emissions Monitoring

Detecting and quantifying emissions during emissions events, especially non-routine flaring or fugitive releases, remains a significant challenge in upstream oil and gas operations. Currently, the market lacks a reliable and cost-effective tool capable of timely monitor and management non-routine emissions events, particularly during short-duration or intermittent events. This gap limits the industry's ability to meet emerging regulatory requirements and emissions reporting standards.

This study investigates the potential of the Virtual Multiphase Mass Flow Meter (vMPFM) to address this monitoring gap. The approach involves comparing gas flow outputs from multiple distributed wellhead vMPFMs with measurements from a centralized physical MPFM installed at the gathering station. Under normal steady-state conditions, the totalized gas flow recorded by the distributed wellhead vMPFMs is expected to match the commingled flow measured at the gathering point. However, during emissions events—especially non-routine occurrences such as fugitive losses—the physical MPFM registered a measurable decrease in gas flow that was not reflected in the upstream wellhead data.

As shown in Figure 10, this discrepancy aligns closely with the timing and magnitude of simulated emissions events, indicating that the combination of virtual and physical MPFMs can effectively detect and localize unaccounted gas losses within the surface piping network, whether due to flaring, venting, or leakage. This hybrid measurement approach enables real-time detection and indirect quantification of emissions and represents a practical, scalable solution for enhancing emissions transparency in upstream operations. The results demonstrate the potential of vMPFM technology to fill a critical market need for continuous, field-deployable emissions monitoring.

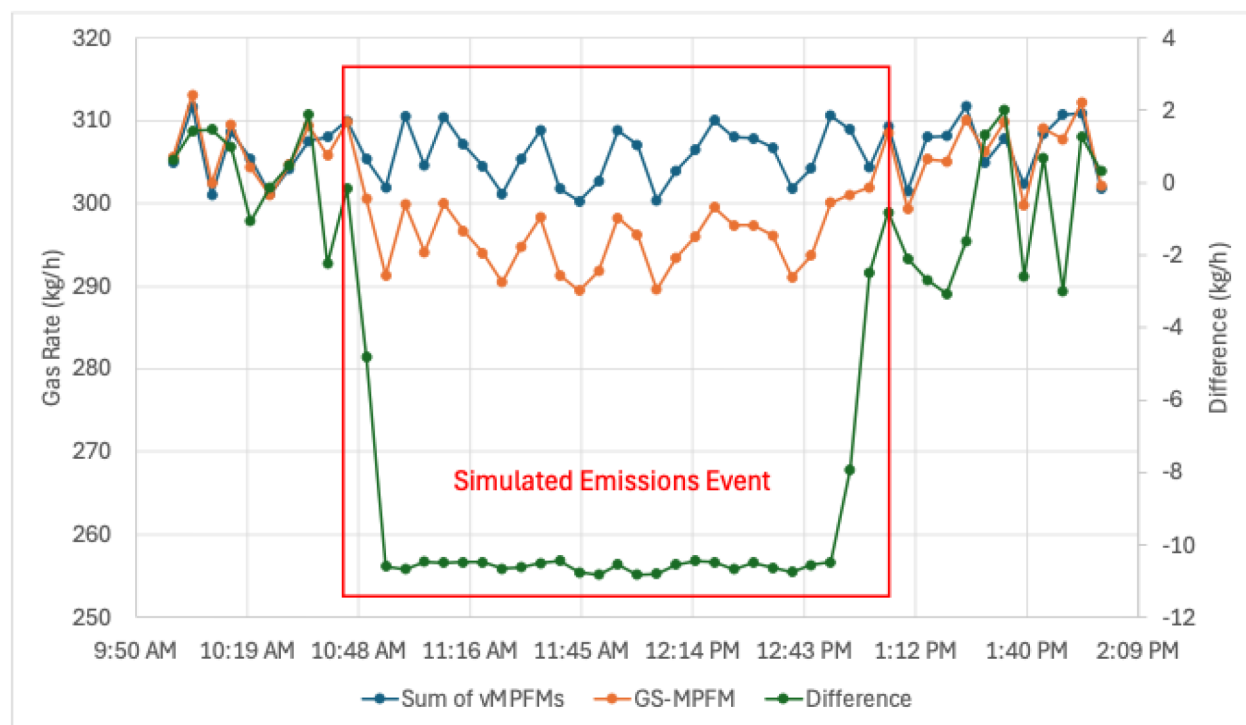


Figure 10—Simulated Emissions Event Captured by Hybrid System (Distributed vMPFMs + Gathering Station MPFM)

This study was conducted under predefined or known conditions, and the vMPFM's performance in more complex, real-world environments with variable flow regimes and multipoint leaks remains to be demonstrated. While mass balance offers a practical approach to detecting unaccounted gas losses, its effectiveness is limited in highly dynamic or phase-slip-dominated multiphase flow conditions. In such cases, external sensors and complementary technologies—such as gas imaging, pressure monitoring, or

tracer-based methods—may be needed to improve accuracy and validate emissions estimates. Future work should focus on field-scale validation, model refinement for transient multiphase behavior, and integration with external sensing systems and automated diagnostics to enhance reliability and regulatory alignment.

vMPFM Application in Leak Detection

To evaluate the potential of the virtual multiphase mass flow meter (vMPFM) to detect and approximate pipeline leaks, a controlled leak test was conducted by inducing artificial leaks along a dedicated section of the flow line. The leakage rate was regulated using a calibrated ball valve positioned downstream of one vMPFM but upstream of the other vMPFM. By adjusting the valve opening to predefined levels, we were able to simulate a range of leak intensities under controlled conditions.

Figure 11 presents the results of the variable valve opening tests. Each test case represents a different leak magnitude, corresponding to specific valve positions ranging from minimal leakage (nearly closed) to significant flow diversion (partially open). These scenarios were designed to reflect realistic field conditions where small to moderate leaks might occur due to equipment failure, corrosion, or valve malfunction.

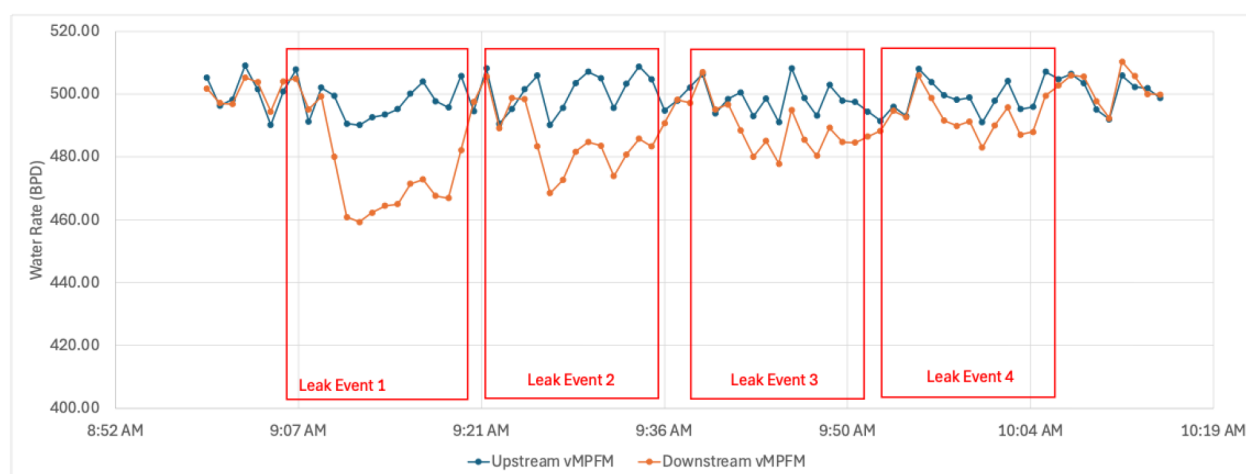


Figure 11—Measured Water Rates by Upstream vMPFM and Downstream vMPFM

During each leak event, a consistent and immediate discrepancy was observed between the water rates measured by the upstream vMPFM and those recorded at the downstream vMPFM. The onset of the leak was detected in real time through this differential, validating the vMPFM system's ability to identify anomalous mass loss events along the pipeline.

To further evaluate the vMPFM-based leak detection approach, the leaked water was independently measured using a calibrated turbine meter installed on the leakage line connected to the ball valve. As shown in Figure 12, the differential flow calculated by subtracting the downstream vMPFM reading from the totalized upstream measurements showed promising agreement with the turbine meter data across all tested leak scenarios.

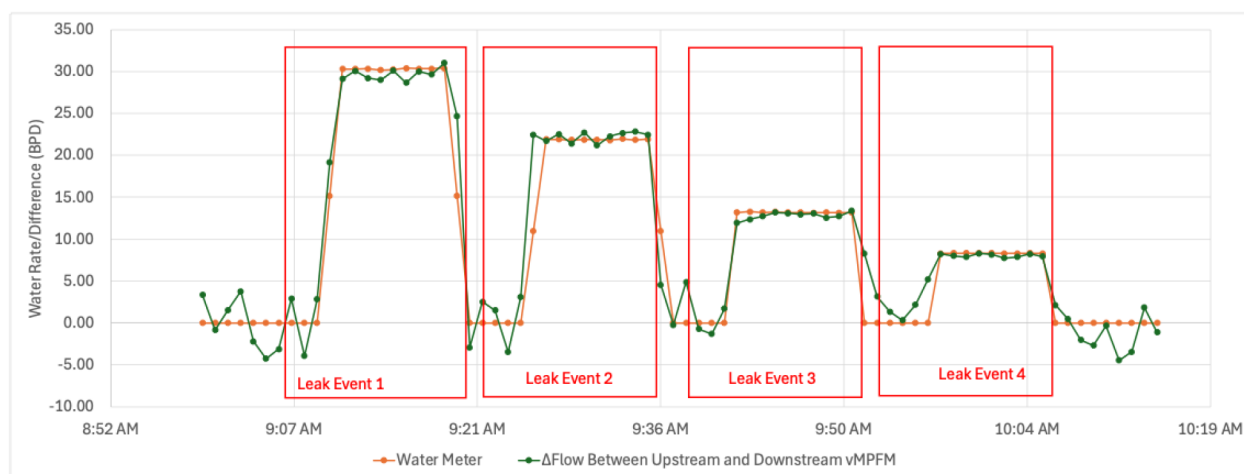


Figure 12—Leak Rates Measured by Water Meter vs. Δ Flow Between Upstream and Downstream vMPFM

These results confirm that vMPFM is capable of not only detecting the occurrence of a leak event with minimal response time but also providing a reliable estimation of the leak rate by applying mass balance principles across the measurement points. This performance illustrates the value of deploying spatially distributed vMPFMs across production infrastructure—not only for flow assurance and emissions quantification but also as a scalable solution for leak detection and localization.

Future research on the application of vMPFM for leak detection will focus on several critical development areas. These include quantifying the minimum detectable leak thresholds under varying flow regimes and sensor configurations, assessing the false positive and false negative rates under realistic field noise and operational transients, and enhancing the model's ability to localize leak points along pipelines with greater spatial resolution.

Study of vMPFM integration with edge computing platforms for autonomous alerting or control actions would represent a valuable contribution to digital field development.

Conclusion

This study introduced and validated an innovative Virtual Multiphase Mass Flow Meter (vMPFM), a system that leverages Physics-Informed Neural Networks (PINNs) and computational fluid dynamics (CFD)-generated synthetic datasets to significantly enhance the accuracy, scalability, and adaptability of multiphase mass flow measurement in the oil and gas industry. Unlike traditional Multiphase Flow Meters (MPFMs), which typically rely on volumetric phase measurements followed by indirect density-based conversions to estimate mass flow rates, the vMPFM directly models and outputs mass flow rates for oil, water, and gas phases. This fundamental shift to a mass-centric approach ensures higher fidelity and reliability, particularly in dynamic environments where fluid densities, compositions, and flow regimes fluctuate unpredictably. By embedding physical constraints—such as the Navier-Stokes equations, interfacial closure laws, and Euler-Lagrange models—directly into the neural network's training process, the Mass vMPFM bridges the gap between data-driven machine learning and first-principles physics, resulting in a hybrid framework that is both robust and generalizable across a wide spectrum of operational conditions.

The experimental validation presented in this paper underscores the vMPFM's performance, with consistent mass flow measurement accuracies within $\pm 5\%$ demonstrated across diverse operational regimes. These include challenging scenarios such as high gas volume fractions (GVF) up to 90%, varying water cuts from 0% to 100%, transient phenomena like slugging and flow regime transitions, and even non-routine emissions events that often confound conventional systems. Benchmarking against physical flow loop data and traditional MPFMs revealed that the vMPFM not only outperforms empirical-based models

in accuracy but also maintains stability without frequent recalibrations, addressing longstanding issues like calibration drift and sensitivity to unsteady flows. For instance, in high-GVF annular or mist flows, where conventional MPFMs exhibited errors up to 12% due to phase slip and separation, the vMPFM achieved 2–3% accuracy by dynamically adapting through its physics-informed architecture. This level of precision stems from the system's ability to incorporate CFD-simulated datasets that span extensive parameter spaces, supplemented by real-world sensor inputs like pressure, temperature, and differential pressure from existing SCADA systems.

By focusing inherently on mass rather than volume, the vMPFM overcomes several key limitations inherent in conventional MPFM technologies. Traditional systems often introduce uncertainties through assumptions about fluid densities, which must be measured separately or estimated via correlations that degrade under variable pressure, temperature, or compositional changes. In contrast, the vMPFM infers these properties in real-time through its PINN core, eliminating the need for additional hardware like radioactive densitometers and reducing associated safety, logistical, and regulatory burdens. This not only lowers capital and operating expenditures but also minimizes measurement uncertainty, enabling proactive production surveillance and optimization. The integration of PINNs further empowers the system to handle sparse or noisy data effectively, generalizing well beyond training datasets and extrapolating to unseen conditions—a capability that traditional empirical models, such as the drift-flux approach, struggle to achieve without extensive recalibration.

Beyond core metering functions, the vMPFM extends its utility to critical applications in leak detection and emissions quantification, leveraging distributed mass-balance methodologies for enhanced asset integrity and environmental stewardship. In controlled experiments, the system detected small-scale leaks in real-time by identifying discrepancies between upstream and downstream measurements, with differential flow estimates aligning closely (within promising margins) to calibrated turbine meter references. Similarly, for emissions monitoring, comparisons between wellhead vMPFMs and centralized gathering station MPFMs revealed unaccounted gas losses during simulated emissions events, offering a scalable solution for quantifying fugitive releases, venting, and blowdowns. This positions the vMPFM as a versatile tool for complying with stringent regulations, such as the U.S. EPA's Subpart W and Canada's Methane Regulation, while supporting broader decarbonization goals. By providing precise, real-time mass flow data for greenhouse gas events, it facilitates accurate reporting, operational interventions, and methane reduction strategies, ultimately contributing to the industry's net-zero ambitions.

The versatile deployment options of the vMPFM—either as a standalone software layer integrated into existing control systems or embedded in minimal edge computing devices—further amplify its practical impact. This architecture drastically reduces barriers to adoption, making advanced multiphase mass flow measurement accessible to remote wellheads, unmanned platforms, marginal wells, and gathering systems that were previously deemed cost-prohibitive or impractical for monitoring. By utilizing conventional field sensors without requiring new installations, it supports field-wide digitalization at a fraction of the cost of hardware-intensive alternatives, unlocking economic value in low-margin assets and enabling "smart wells" and digital twins for autonomous operations.

Looking ahead, the vMPFM represents an important advancement in the digital transformation of upstream oil and gas operations, aligning seamlessly with global trends toward sustainability, efficiency, and intelligent field development. Its ability to unify physical modeling with AI/ML-driven insights not only addresses immediate challenges in multiphase mass flow measurement but also paves the way for future innovations. Potential extensions could include deeper integration with IoT sensors for predictive maintenance, enhanced anomaly detection through advanced analytics, or coupling with reservoir simulation models for holistic asset management. While this study focused on validation under controlled and simulated conditions, field trials in diverse reservoirs—such as offshore or unconventional plays—will further substantiate its real-world efficacy. Ultimately, by bridging the divide between conventional metering and next-generation digital solutions, the vMPFM empowers operators to achieve

higher operational resilience, reduce environmental footprints, and drive sustainable growth in an increasingly regulated and data-centric energy landscape. This framework not only redefines multiphase flow management but also sets a new standard for AI/ML-enabled technologies in fostering low-carbon, efficient, and intelligent energy systems worldwide.

List of Abbreviations

AI	: Artificial Intelligence
BP	: British Petroleum
BPD	: Barrels per Day
CFD	: Computational Fluid Dynamics
CAPEX	: Capital Expenditure
Δ	: Delta (Difference or variation between two values)
EPA	: Environmental Protection Agency
GS	: Gathering Station
GVF	: Gas Volume Fraction
kg/h	: kilogram per hour
ML	: Machine Learning
MMSCFD	: Million Standard Cubic Feet per Day
MPFM	: Multiphase Flow Meter
PINN	: Physics-Informed Neural Network
SCADA	: Supervisory Control and Data Acquisition
SR-RK-PINN	: Self-Regressive Runge-Kutta Physics-Informed Neural Network
vMPFM	: Virtual Mass Multiphase Flow Meter
WC	: Water Cut
WH	: Well Head

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